

SHIVAM KUMAR

+91-9129085031 | shivamkm07@gmail.com | linkedin.com/in/shivamkm07 | github.com/shivamkm07

SUMMARY

Senior software engineer with 5+ years on Azure Container Apps, building and operating cloud-native distributed systems, AI-powered SRE/agentive platforms, and CNCF open-source infrastructure (Dapr). Work spans distributed-systems design, Kubernetes-based serverless and Functions platforms, large-scale service reliability, and developer-facing APIs across multiple teams and clouds.

Focus: Distributed Systems · Cloud-Native & Kubernetes Platforms · AI/LLM Agentive Systems · Site Reliability Engineering · CI/CD & Release Engineering

EXPERIENCE

Software Development Engineer II

Jul 2024 – Present

Microsoft – Azure Container Apps

Hyderabad, India

- Led the design and delivery of **AI-powered autonomous SRE agents** for ACA — cut P50 incident time-to-mitigate by **97%** (185→6 min) and time-to-resolve by **64%**; sustained **~99% RCA coverage** at **~76% on-call-rated accuracy**, **auto-resolving 57 of ~400 CRIs** and authoring **~132 repair items**. Established the Ev2 release pipeline for the agent.
- Drove **ACA Express Public Preview** readiness across the ACA and **Azure Dev Compute (ADC)** orgs — owned the design and stakeholder sign-off for **ephemeral volume support** and **Log Analytics customer-log delivery**, the P0 migration path for **~100K+ ACA environments** and **~\$40M** projected annual COGs savings.
- Architected the **Foundry Control Plane integration** making ACA a first-class host for agentive apps — designed a new cross-cloud preview API (Control Plane + CRD) with auto-detection via **privileged host-process inspection** and customer-declared modes, covering **~2,000+ agentive apps**.
- Owned **reliability of Azure Functions on ACA** (~5K managed, ~1,322 external apps) — designed **graceful drain** that retired **3 recurring CRIs**, plus queueLength/scale-rule overrides, JWT clock-skew leeway, function-keys management, and invocation tracing; migrated telemetry to **central Kusto clusters**, unblocking alerting and agent skills. Mentored a junior engineer to co-own the feature set.
- Delivered **Session Management data-plane APIs** and resolved critical async-execution bugs for ACA Sessions (~300 external customers); added .NET/Node.js code-interpreter support **adopted by the Security Copilot team**.
- Led **Dapr AKS Extension** rollouts (v1.15/v1.16) across all Azure regions with daily artifact refresh and CVE remediation over **~400 clusters**; integrated **Azure Key Vault** with the Managed Environment Storage API, securing **~27K daily calls** by eliminating raw connection strings.
- Served as **on-call DRI** across rotations, resolving **240+ incidents (65+ CRIs, 36 Sev2 LSIs)** well under platform baselines; fed incident learnings back as agent skills and repair items, and **mentored 2 new hires** through their first on-call rotations.

Software Development Engineer

Jul 2021 – Jul 2024

Microsoft – Azure Container Apps / Dapr

Hyderabad, India

- Drove core **Dapr** (CNCF) components to stable — MQTT PubSub, CosmosDB, Redis/PostgreSQL state & config stores, and the Configuration API; implemented **cascade terminate/purge** for Dapr Workflows across runtime and SDKs; added bulk subscribe for Azure Event Hubs.
- Diagnosed and fixed a critical **Dapr Workflows scalability defect** (workflows stalling beyond two instances) through deep **DurableTask-framework** debugging, adding metrics and logging — a widely-cited reliability win for production workflow customers.
- Architected and solo-owned the **internal Dapr fork strategy** across 5 repositories — built a Git patch-based framework, repo structure, and CI/CD from scratch, automating OSS-to-internal releases within a **<1-day SLA** and producing multi-arch images (ARM64/ARM/AMD64) for all supported Dapr versions from a **single pipeline**.
- Enabled **airgap installation**, winget, and MSI for the Dapr CLI; added Mariner/distroless images and handled 1P Container Registry migration; consolidated builds into **OneBranch-compliant** pipelines with Ev2 deployment.
- Helped launch **ACA Sessions public preview** — added Node.js code-interpreter support and resolved critical Python/Node.js session bugs surfacing in production.

Quantitative Research Intern

May 2020 – Jul 2020

JP Morgan & Chase – E-Trading

Mumbai, India (Remote)

- Built price prediction models using **LSTM**, feedforward neural networks, and regression with univariate feature selection; improved predictive power by **40%** for HK names through feature standardization.
- Implemented L1 and L2 features for optimizing model parameterization across multiple asset classes.

TECHNICAL SKILLS

Languages: Go, Python, C#/.NET, JavaScript/TypeScript, C/C++

Cloud & Infrastructure: Azure (Container Apps, AKS, Functions, Key Vault, Event Hubs, CosmosDB), Kubernetes, Docker, Dapr (CNCF), Helm

Tools & CI/CD: OneBranch (1ES), Ev2, Azure DevOps Pipelines, GitHub Actions, Git, Geneva, Kusto/ADX, CodeQL, KEDA, Argus

Domains: Distributed Systems, AI/LLM Agents, AI-Augmented Engineering (Copilot review skills, e2e stamp validation), Site Reliability (SRE), CI/CD & Release Engineering

RESEARCH

Data Race Detection for Task-Parallel Programs

Jun 2019 – Aug 2021

Supervisor: Prof. Swarnendu Biswas, IIT Kanpur

- Designed novel algorithm **Tasker** integrating vector clocks with tree-based techniques; implemented using **LLVM** pass for memory instrumentation on task-parallel programs.
- Achieved **1.48X speedup** over PTRacer baseline on 128GB Intel Xeon system with lower memory overhead.
- “Efficient Data Race Detection of Structured Task Parallel Programs Using Vector Clocks” — S. Kumar, A. Agrawal, S. Biswas | Submitted to ISCGO 2022.

EDUCATION

Indian Institute of Technology, Kanpur

CGPA: 9.4/10.0

Bachelor of Technology in Computer Science and Engineering

2017 – 2021

ACHIEVEMENTS & LEADERSHIP

RockStar Award, Microsoft ('22, '23) for exceptional contributions | **Academic Excellence Award** (×2), IIT Kanpur

JEE Advanced **AIR 715** | JEE Mains **AIR 348** | KVPY Scholar | NSE Physics **Top 1%** | Ram Prakash Chopra Scholarship

Academic Mentor, IIT Kanpur ('18-19) | Secretary, Dramatics Club, IIT Kanpur ('18-19)